OBD2 sniffer: revision A

A read-only CAN and OBD2 scanner. Powered from the laptop's USB port only.
Design notes and requirements: docs/design/circuit-board.md, docs/design/requirements.md.
Parts, datasheets and the pre-order checklist: hardware/rev-a/README.md.

Sheets

USB and power



File: usb-power.kicad_sch

USB-C socket, fuse, ESD protection,
3.3 V regulator, PWR light.

Microcontroller



File: microcontroller.kicad_sch

STM32F446RET6, crystal, decoupling, reset and boot,
SWD and debug serial headers, BUS light.

CAN



File: can.kicad_sch

Two CAN transceivers, LISTEN/OBD switch and the silent signal,
TX light, CAR/TEST jumpers, termination, car connector, test points.

Sheet: /		
File: obd2-sniffer-rev-a.kicad_sch		
Title: OBD2 sniffer, revision A		
Size: A4	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/4

USB and power

The board is powered from the laptop's USB port only (PWR-5).
The car's 12 V supply has no connection to the board (PWR-8).

USB-C socket

5.1 k on each CC pin: the host switches its 5 V on.

Fuse

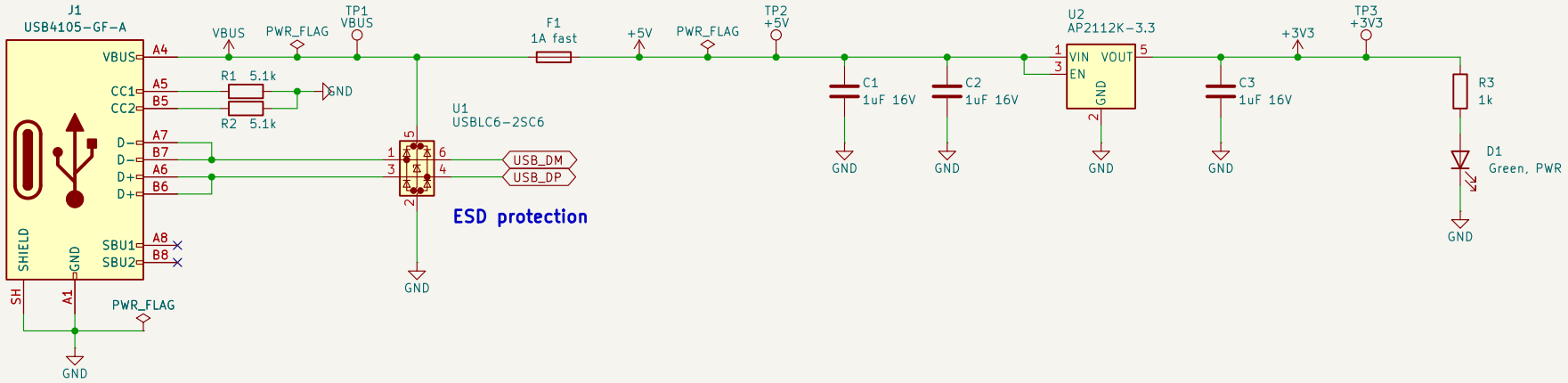
1 A fast, 0.13 ohm: a short on the board can't draw a dangerous current.

3.3 V regulator

1 uF ceramic in and out, as the datasheet asks. Enable is tied to the input: always on.

PWR light

On whenever 3.3 V is present, with or without firmware (USE-3).



The socket's shield is tied to ground. SBU pins are not used.

Total capacitance on the USB 5 V stays under 10 uF, the limit a USB device may present when plugged in.
The transceivers take the 5 V directly and need at least 4.5 V: the fuse drops about 35 mV at 250 mA.

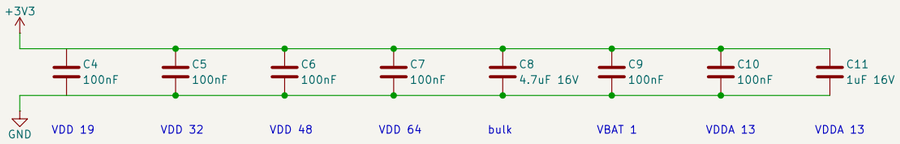
Sheet: /USB and power/		
File: usb-power.kicad_sch		
Title: USB and power		
Size: A4	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/4

Microcontroller

STM32F446RET6 in LQFP64, the same chip as the development board.
Supply pins, capacitors and boot pins follow ST's datasheet DS10693 and application notes AN4488 and AN2606.

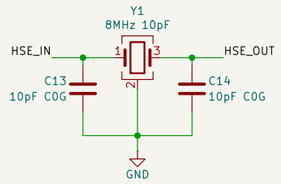
Decoupling

100 nF at each supply pin, placed at the pin. One 4.7 uF for the chip.
VDDA joins 3.3 V directly: the analogue converter isn't used.



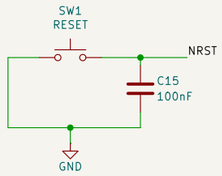
8 MHz crystal

ECS ECX-53B, 10 pF load, at most 100 ohm.
Load capacitors: 2 x (10 pF − 5 pF stray) = 10 pF.
USB and the USB bootloader need this clock.



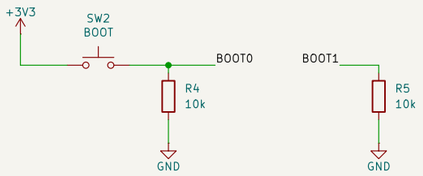
Reset

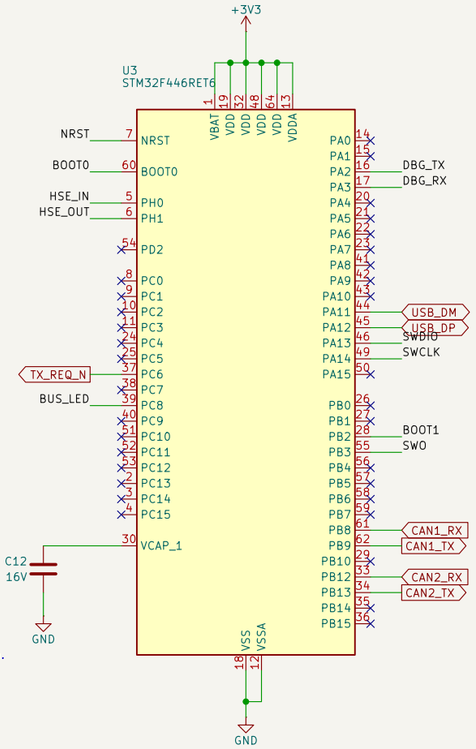
The chip has its own pull-up on NRST.



Boot

Hold BOOT while pressing RESET to start the chip's USB bootloader (USE-4).
BOOT0 is low otherwise, and BOOT1 (PB2) is always low, as the bootloader needs.

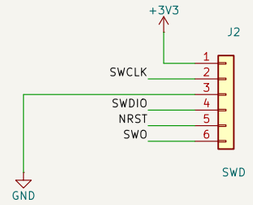




One VCAP pin on LQFP64:
a single 4.7 uF, ESR under 1 ohm.

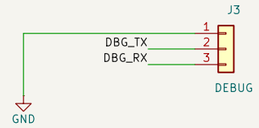
SWD debug header

For programming and debugging.
Pin 1 lets the debugger sense 3.3 V.



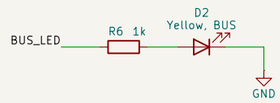
Debug serial header

Start-up and crash reports, 3.3 V levels.



BUS light

Driven by firmware (USE-3).



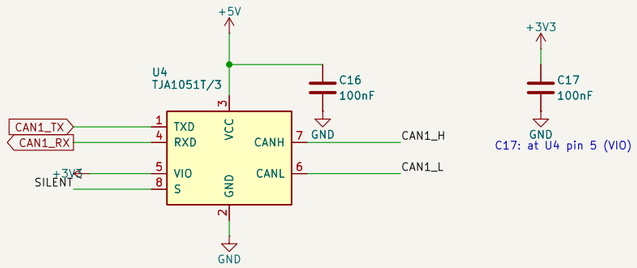
Sheet: /Microcontroller/		
File: microcontroller.kicad_sch		
Title: Microcontroller		
Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/4

CAN

CAN1 is the car's bus. CAN2 exists only for the self-test (CAN-7).
The safety rules for this sheet are in docs/design/safety.md and in the checklist in hardware/rev-a/README.md.

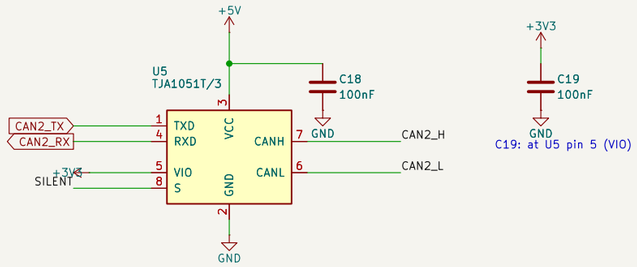
CAN1 transceiver: the car

Bus pins withstand -58 to +58 V. Transmit timeout 0.3 to 5 ms.
Unpowered, it disconnects from the bus (PWR-8).



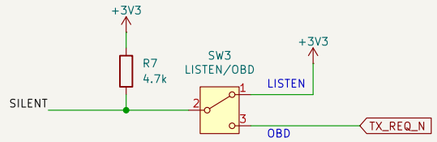
CAN2 transceiver: self-test only

Reaches CAN1 only through the TEST position of the jumpers.



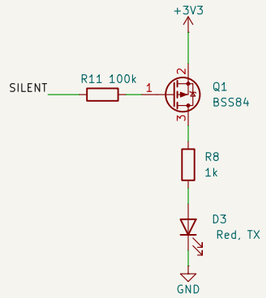
LISTEN/OBD switch and the silent signal (SAF-3)

SILENT high: both transmitters are off. SILENT low: they can transmit.
Left open, the transceiver's own pull-down would select 'can transmit', so R7 pulls SILENT high:
after a reset, a crash or with a blank chip, PC6 floats and the board is silent.
LISTEN ties SILENT to 3.3 V, whatever the firmware does. OBD hands it to the firmware (TX_REQ_N, PC6).



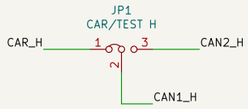
TX light (USE-3)

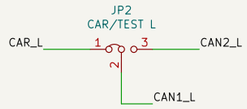
Switched by SILENT itself, not by firmware.
Lit only while the board is able to transmit.
R11 keeps a fault in the transistor
from pulling SILENT low.



CAR/TEST jumpers

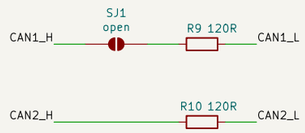
Two 3-pin headers, one for each bus wire. A jumper cap joins the middle pin (CAN1) to one side:
pins 1-2 CAR: CAN1 to the car connector. pins 2-3 TEST: CAN1 to CAN2.
No position joins CAN2 to the car connector. The headers sit apart, so a cap can't bridge between them.





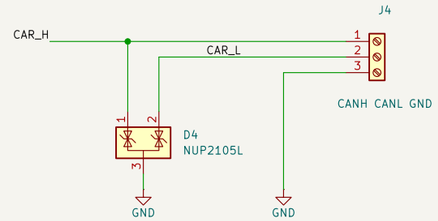
Termination

CAN1: only if SJ1 is bridged with solder. It is open as built, because the car's bus
has its own termination and a third resistor would change it (CAN-6).
CAN2: always fitted. It terminates the self-test bus.



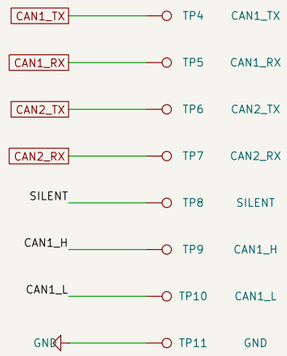
Car connector

Three terminals: CAN high, CAN low, ground.
There is no terminal for the car's 12 V (OBD2 pin 16).
OBD2 pin 6 to CANH, pin 14 to CANL, pins 4 and 5 to GND.



ESD protection: working voltage 24 V,
so a short to the car's battery
doesn't make it conduct.

Test points



Sheet: /CAN/		
File: can.kicad_sch		
Title: CAN		
Size: A3	Date: 2026-11-11	**Rev: A**
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